downstream movement with the pulse is accompanied by energy depletion. Rumiantsev et al. reported an average plasma energy density merely 0.35 kJ/cm^3 for breakdown in water by a powerful 325-µJ, 1240-nm, 170-fs pulse focused at $NA = 0.5$ [219]. Due to the pulse energy far above threshold, the plasma length amounted to 210 µm, while the pulse itself was only 40 µm long. Therefore, the breakdown wave moved with the pulse and its energy was depleted during propagation even at large NA.

5.4. Plasma structure, energy density and pressure in nanosecond breakdown

Luminescent ns plasmas are characterized by an intricate substructure within the luminescent region and by a large size going far beyond the laser cone angle at large NA (Figs. 7 and 8). We will see that both features are linked by radiative energy transport from the regions of primary energy deposition and reabsorption in the surrounding liquid, which inflates the plasma and reduces the average energy density.

The finding that the average energy density in high-density ns and fs plasmas is similar even though the pulse energies in ns breakdown are much larger (Table 3) is surprising at first sight. Femtosecond breakdown produces merely one set of free electrons, with thermalization after the laser pulse. By contrast, the cycle of free-electron generation and thermalization is repeated many times during a ns pulse. Therefore, one would expect much larger energy densities in ns breakdown than with fs pulses. However, fs plasmas are confined to the cone angle of the pump laser beam, whereas luminescent plasmas produced by tightly focused ns pulses extend well beyond the irradiated cone (Fig. 7). They feature an inhomogeneous core region of primary energy deposition that is surrounded by a diffuse luminescent halo. The region of primary energy deposition by 532-nm ns pulses is demarcated by branched plasma strings that scatter the incoming laser light [Figs. 7(c),(e),(f)]. While the energy density averaged over the entire diffuse plasma luminescence is 14 kJcm^{-3}, the average over primary energy deposition zone within the laser cone angle is 81 kJ/cm^3. Peak values are likely much higher because the strings occupy less than 1/5 of the core volume. This line of reasoning suggests transient local energy densities of up to 400 kJ/cm^3. Similar inhomogeneities as those produced by 532-nm pulses likely exist also in UV and IR ns-laser-produced plasmas but are not visible on the photographs because the digital camera used in the experiments does not record the scattered UV and IR laser radiation.

What causes the inhomogeneous energy deposition in nanosecond breakdown leading to strings in the region of primary energy deposition? The inhomogeneities cannot originate from nonlinear beam propagation because at $E_L = 670$ µJ [Figs. 7(e) and (f)] the laser peak power remains more than 26 times below the threshold for catastrophic self-focusing in water, which is $\approx 2\times10^6$W at 532 nm [263]. Instead,

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they result from an inherent instability of avalanche ionization that is supported by the large number of doubling sequences in ns breakdown. Since the avalanche ionization rate is proportional to irradiance and to the local free-electron density ρ_c, the avalanche will grow faster at locations where ρ_c is already larger than elsewhere. Small intensity fluctuations across the pump laser beam result in larger fluctuations of the seed electron density produced by PI, which are then further enhanced by AI $\propto \rho_c$. Strings are produced, when the zones of high electron density grow upstream as the breakdown wave propagates towards the incoming laser beam [108,133,134]. The speed of the upstream moving breakdown wave during ns breakdown is several orders of magnitude slower than with fs pulses [the average speed is ≈ 9.1 km/s for the 733-µJ ns pulse in Fig. 7(c), compared to 28570 km/s for the 0.57-µJ fs pulse in Fig. 7(a). That provides sufficient time for a large number of doubling sequences in the ionization avalanche.

The small diameter (≈ 1 µm) and the large surface of the branched filamentous regions allow for efficient laser-plasma coupling [138,151,236,238,264]. Above ρ_{crit}, about 30% of the pump light are absorbed within less than 1 µm, and the rest is reflected or backscattered [236,257]. The increase of plasma absorptivity and reflectivity, when ρ_c exceeds ρ_{crit}, [151,256], enhances the instabilities promoting string formation. The locally increased absorptivity results in a runaway towards very high ionization levels followed by rapid heating of the free electrons, and the increased reflectivity promotes the upstream growth of the string tips. Local inhomogeneities may lead to branching, such that a fractal high-density plasma structure evolves.

The strings in Figs. 7(e) and (f) resemble streamers in DC breakdown in gases [265], water [266-271], and lightning [272] – but on a much smaller spatial scale. We use the name "string" rather than "streamer" to emphasize the difference between AC and DC breakdown. Nevertheless, there are similarities too. The strong DC electric field emanating from streamer tips also accelerates electrons and drives AI, which then results in streamer growth [266-268,273]. For positive streamers, seed electrons are produced through photoionization by energetic photons emitted from the tips [265]. Their propagation velocity is > 20 km/s [266,267], similar to the speed of the breakdown front in nanosecond optical breakdown. However, since a high ionization level is reached in the highly absorbing strings, the energy density is much higher than in DC streamers before spark formation. Both strings and streamers exhibit a tree-like fractal structure [265,274,275]. The exploration of branching mechanisms and of the fractal character of branching trees is an active field of research [276-278].

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String formation arising from an intrinsic AI instability in conjunction with the upstream movement of the optical breakdown wave differs from multiple filament formation arising from instabilities of nonlinear light propagation during the downstream movement of an energetic laser pulse [204,252,279]. Both string and filament formation occur during single shots, unlike laser-induced periodic structures at surfaces and periodic structures inside bulk media that arise from a cumulative interplay between electric field and thermomechanical effects during multiple exposures [280-283].

Electron diffusion and energy transport by energetic photons are potential mechanisms for the generation of the large luminescent halo within and around the region of primary energy deposition. Figure 14 compares their possible range as a function of electron or photon energy, respectively.

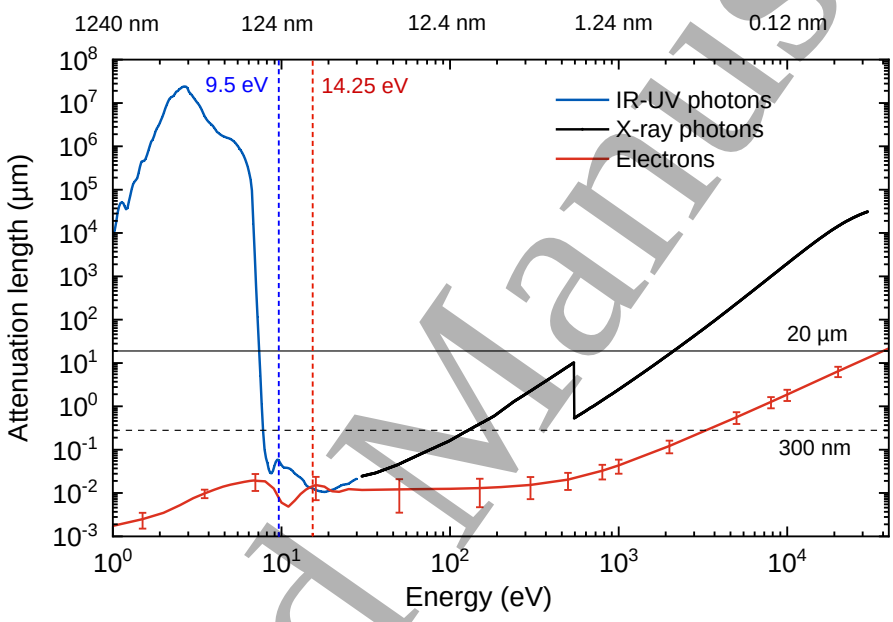

Figure 14 Energy dependence of the attenuation length of photons and electrons in water. Data are taken from [284,285] for photons of IR to UV wavelengths, from [286,287] for x-rays, and from [288] for the attenuation length of electron beams. The latter values were obtained through Monte Carlo simulations, and the vertical bars indicate 95% confidence interval [288]. Photons can produce CB electrons by one-photon ionization, when their energy is larger than the band gap of water (E_{gap} = 9.5 eV), and electrons can produce CB electrons, when their energy is larger than the impact ionization energy ($1.5\times E_{gap}$ = 14.25 eV) [37]. These energy levels are marked by vertical dashed lines. The penetration depth of an electron beam provides an upper limit for the diffusion length of electrons from a plasma, where their motility is limited by the attraction from positive charges. The dashed horizontal line demarcates an attenuation length of 300 nm for which photons emitted from high-density strings of about 1 µm diameter are reabsorbed within the string region. This applies to photons in the energy range between 7.7 eV and 128 eV. The transport distance needed to form a luminescent halo extending 20 µm beyond the laser cone angle is marked by a solid line. Photons with energies between 128 eV and 2.16 keV have penetration depths between 300 nm and 20 µm; large enough to form the observed halo. The drop of penetration depth at 532 eV is due to the K-absorption edge of oxygen [286].

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